

Results

-> Real-time cancer risk prediction screenshot showing user feedback and dynamic model retraining.

```
===== Real-Time Cancer Risk Prediction =====
Enter Age: 20
Enter Gender: 0
Enter BMI: 16.4
Enter Smoking: 1
Enter GeneticRisk: 0
Enter PhysicalActivity: 0
Enter AlcoholIntake: 1
Enter CancerHistory: 0
Enter DietQuality: 0
Enter RadiationExposure: 0
Enter OccupationRisk: 0
Enter StressLevel: 0
Enter HormonalTherapy: 0
Enter PreExistingConditions: 0
Enter EnvironmentalExposure: 0
Enter SleepDuration: 3
Enter AirPollutionExposure: 0
Enter ExerciseType: 1
Enter ProcessedMeatIntake: 1
Enter SugarConsumption: 0
Enter Region: 0
Enter EducationLevel: 0
1/1 ██████████ 0s 46ms/step

Prediction: Low Risk
What is the actual diagnosis? (0 = No Cancer, 1 = Cancer Diagnosed): 0
Actual diagnosis recorded. Updating models...

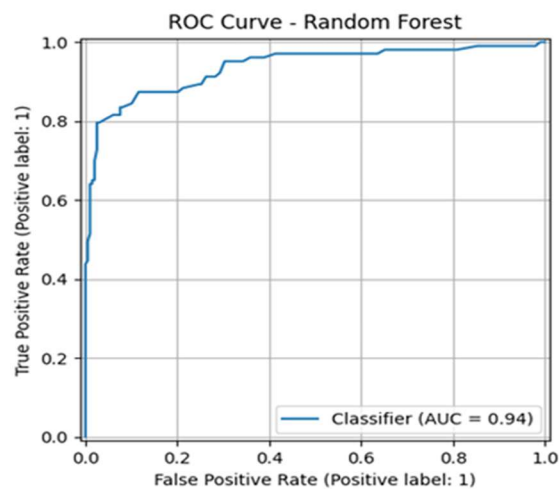
Do you want to predict for another person? (yes/no): 0
Dataset updated and saved successfully as 'Enhanced_Cancer_Data_Updated1.csv'!
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-> Random Forest Classifier Performance

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--- Random Forest ---
              precision    recall  f1-score   support

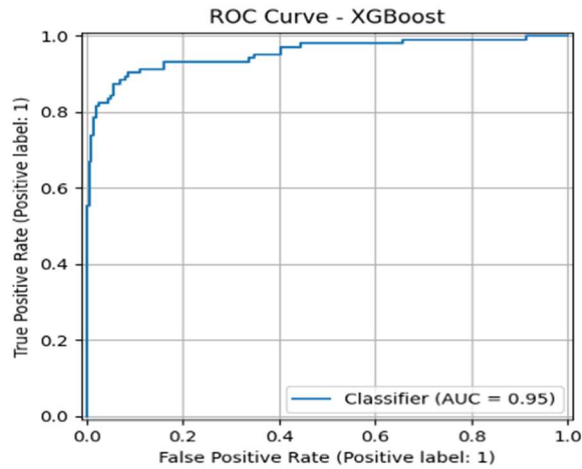
     0       0.88      0.97      0.93       198
     1       0.94      0.75      0.83       103

 accuracy          0.90       0.90      0.90       301
 macro avg         0.91      0.86      0.88       301
 weighted avg      0.90      0.90      0.89       301
```



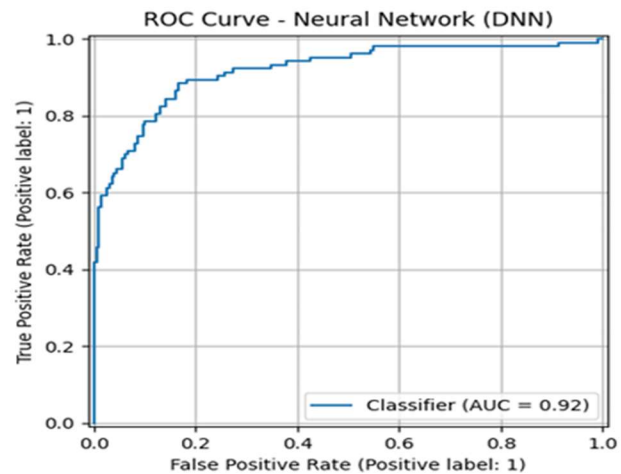
-> XGBoost Classifier Performance

--- XGBoost ---				
	precision	recall	f1-score	support
0	0.91	0.96	0.94	198
1	0.92	0.83	0.87	103
accuracy			0.92	301
macro avg	0.92	0.89	0.91	301
weighted avg	0.92	0.92	0.92	301



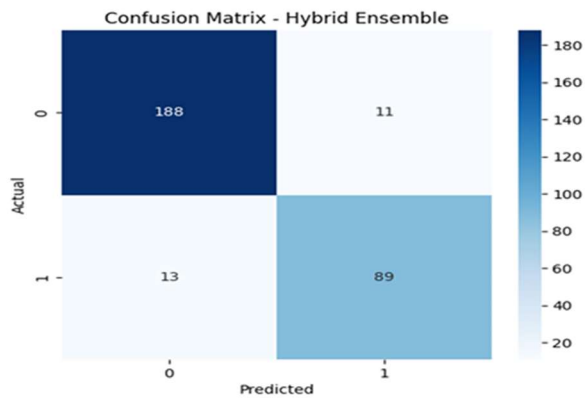
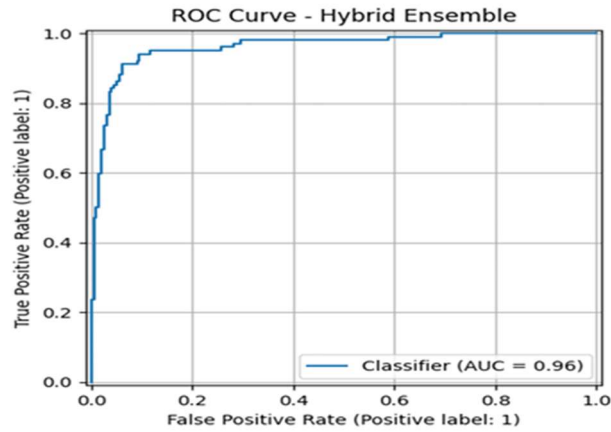
-> Deep Neural Network (DNN) Performance

--- Neural Network (DNN) ---				
	precision	recall	f1-score	support
0	0.89	0.91	0.90	198
1	0.83	0.79	0.81	103
accuracy			0.87	301
macro avg	0.86	0.85	0.85	301
weighted avg	0.87	0.87	0.87	301



-> Hybrid Ensemble Model Evaluation

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=== Hybrid Ensemble Model Accuracy ===  
Train Accuracy: 0.99  
Test Accuracy: 0.91  
✅ The Hybrid Ensemble Model is Properly Fitted.
```



```
Classification Report:  
              precision    recall  f1-score   support  
  
     0       0.94       0.94       0.94       199  
     1       0.89       0.87       0.88       102  
  
    accuracy       0.92       0.92       0.92       301  
   macro avg       0.91       0.91       0.91       301  
  weighted avg       0.92       0.92       0.92       301  
  
ROC-AUC Score: 0.96
```